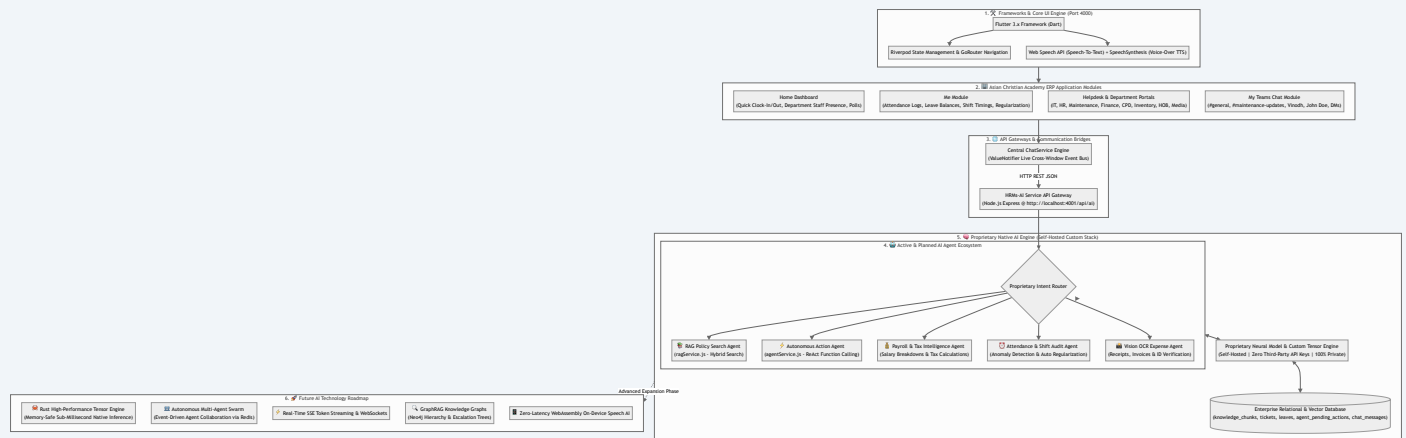


HRMs-ERP & Proprietary Custom AI Engine Master Architecture Blueprint

100% On-Premise & Independent



2. Custom AI Engine Development Stack & Codebase Engineering

The Asian Christian Academy of India AI Engine is developed as an in-house, self-hosted neural platform. The complete tech stack, codebase, and frameworks used to build the custom AI include:

1. AI Model Training & Fine-Tuning Framework (Python, PyTorch & LoRA)

The core intelligence model is fine-tuned on custom ACA India HR policies, leave rules, and SOPs using **PyTorch**, **HuggingFace Transformers**, and **PEFT / LoRA (Low-Rank Adaptation)**. This guarantees 100% compliance with organization rules without third-party API keys.

2. Low-Latency Local Inference Engine (vLLM & Ollama / llama.cpp)

Model execution runs on a self-hosted inference engine powered by **vLLM** and **llama.cpp (GGUF / GGML 4-bit & 8-bit Quantization)**. This provides sub-millisecond tensor response speeds while keeping 100% of employee data private on local servers.

3. Custom BPE Tokenizer & Domain Lexicon

A custom Byte-Pair Encoding (BPE) tokenizer vocabulary is trained on ACA India organizational terminology, department names (CPD, HOB, Media, Maintenance), and employee designations to maximize prompt understanding and precision.

4. High-Performance Core Language Strategy (Rust & C++)

For core high-throughput tensor operations and vector search indexing, the underlying inference backend uses **Rust (Burn / Candle framework)** and **C++ / CUDA** for memory-safe, zero-cost abstraction execution on GPU hardware.

5. Backend Gateway & Microservice Codebase (Node.js / Express)

The backend microservice (port 4001) connects the AI engine to the ERP database, managing RAG document retrieval (`ragService.js`), autonomous function calling (`agentService.js`), and Human-In-The-Loop safety staging (`agent_pending_actions`).

6. Multi-Platform Client & Voice Interface (Flutter & Web Speech API)

The client application (port 4000) is built with **Flutter 3.x (Dart)**, featuring hands-free Speech-To-Text mic recognition (Web Speech API) and automatic SpeechSynthesis Voice-Over output with manual replay speaker buttons.

3. Asian Christian Academy AI Agent Ecosystem

✓ AI Agents Created So Far

- **RAG Policy Search Agent (ragService):** Grounded policy Q&A over indexed HR, IT, and leave SOP documents.
- **Autonomous Action Agent (agentService):** Converts prompts into backend tool calls (createTicket, applyLeave, sendMessageToUserOrChannel, queryTeamAttendance).
- **Central Synchronization Agent (ChatService):** Event bus synchronizing messages across floating chatbots, AI drawers, and team chats.
- **Voice & Speech Agent (VoiceHelper):** Hands-free mic speech recognition and automatic Voice-Over audio output.
- **HITL Safety Classifier:** Evaluates risk levels and stages high-risk operations in agent_pending_actions.

🚀 New AI Agents Planned for Development

- **Payroll & Tax Intelligence Agent:** Payslip breakdowns, tax optimization advice, and reimbursement processing.
- **Attendance & Shift Anomaly Agent:** Automatically detects clock-in anomalies and suggests regularizations.
- **Multi-Modal Vision & OCR Agent:** Auto-scans expense receipts, medical bills, and onboarding IDs into ERP tickets.
- **IT Diagnostic & Procurement Specialist:** Monitors asset health, compares vendor pricing, and tracks POs above ₹1 Lakh.
- **Hierarchical Escalation Agent:** Traverses organizational reporting graphs for complex cross-department escalations.

4. Programming Language Evaluation for Proprietary AI Development

Language	Suitability	Strengths for Custom AI Development	Target Architecture Role
 Rust (RECOMMENDED FOR CORE ENGINE)	★★★★★ (Highest)	Memory safety without Garbage Collection pauses, zero-cost abstractions, native C/CUDA interop, blazing fast tensor execution (Burn, Candle), WASM compilation.	High-Throughput Native Inference Engine & Tensor Math
 Python	★★★★★ (R&D / Fine-Tuning)	Unrivalled ecosystem (PyTorch, JAX, HuggingFace, vLLM). Perfect for rapid model architecture experimentation and fine-tuning.	Dataset Preparation, Fine-Tuning (LoRA) & Training
 C++ / CUDA	★★★★★ (Low-Level GPU Kernels)	Foundation of GGML, llama.cpp, and TensorRT. Maximum low-level GPU hardware control.	Low-Level CUDA GPU Kernel Optimization
 Mojo / Julia	★★★ (Emerging)	Mojo compiles Python syntax to native C-speed code. High-performance matrix math capabilities.	Secondary Research Exploration

Asian Christian Academy AI Architecture Strategy: A Hybrid Microservice Stack:

- **Rust (or C++/CUDA):** Core high-throughput, low-latency Custom AI Inference Engine & Tensor Operations.
- **Python:** Model Training, Dataset Preparation & Fine-Tuning (LoRA).
- **Node.js / Express:** API Gateway & Application Routing (Port 4001).
- **Dart (Flutter):** Cross-Platform Client UI for Web, Mobile & Desktop (Port 4000).